Name: Muhammad Junaid Umar **Section: CS (C)**

LaB-1

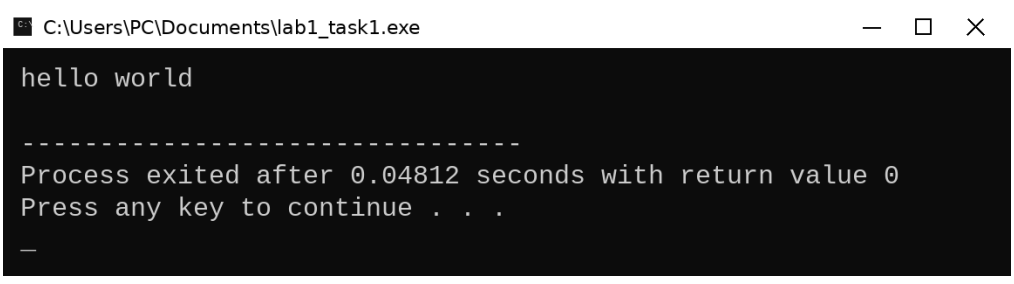
Task-1

Run the following program to get output input

```
#include <iostream>
using namespace std;
int main()
cout << "hello world" << endl;
system ("Pause");
return 0;
}
```

output

=> Hello world



Task 2

write a c++ program to print the following lines

You are 10 year's old
you are too young to play the game

```
#include <iostream>
using namespace std;
int main()
{
cout << "You are 10 year old" <<endl;
cout <<"you are too young to play the game" <<endl;
system ("pause");
return 0;
}
```

You are 10 Year old
You are too young to play the game

```
C:\Users\PC\Documents\lab1_task2.exe
You are 10 Year old
You are too young to play the game

--------------------------------
Process exited after 0.05197 seconds with return value 0
Press any key to continue . . .
```

Task :- 3

Star pattern

```
#include <iostream>
using namespace std;
int main()
{
cout << "*" << endl;
cout << "**" << endl;
cout << "***" << endl;
cout << "****" << endl;
cout << "*****" << endl;
return 0;
```

output

```
*
**
***
****
*****
```



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PF. Lab-2

Task-1

Run these programs and observe output

program 1

```
#include <iostream>
using namespace std;
int main()
{
char letter;
letter = 'A';
cout << letter << '\n';
letter = 'B';
cout << letter << '\n';
cin >> letter;
return 0;
}
```

(output)

A

B



Program-2

[input]

```
# include <iostream>
using namespace std;
int main ()
{
char letter;
letter = 65;
cout << letter << '\n';
letter = 66;
cout << Letter << '\n';
cin >> letter;
return 0;
}
```

A

B

C:\Users\PC\Documents\lab2_task1_p2.exe

```
A
B
_
```

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Program-3

[input]

```
# include <iostream>
using namespace std;
int main ()
{
int letter;
letter = 'A';
cout << letter << "\n";
letter = 'B';
cout << letter << '\n';
cin >> letter;
return 0;
}
```

[output]

```
65
66
```



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Task-2

```
# include <iostream>
using namespace std;
int main ()
{
 int a,b;
 cout << " Enter two number:";
 cin >>a >>b;
 cout << "Addition:" << a+b <<endl;
 cout << " Substraction:" << a-b <<endl;
cout << "Multiplication:" << axb <<endl;
 cout << " Divison:" << a/b <<endl;
 cout << "Modulas:" << a%b <<
 endl;
  system ("pause");
 return 0;
}
```

output (if a=17, b=5)

enter two number 17 5
Addition: 22
Substration = 12
Multiplication = 85
Division = 3
Modulas = 2

```
C:\Users\PC\Documents\lab2_task2.exe
Enter two number:17 5
Addition: 22
Substraction: 12
Multiplication: 85
Divison: 3
Modulas: 2

--------------------------------
Process exited after 8.742 seconds with return value 0
Press any key to continue . . .
```

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Task - 3

write output of following code

```
# include "stdafx-h"
# include <iostream>
using namespace std;
 int main ()
{
 int val1;
 int val2;
 Ind val3;
cout << "Please enter your 3 numbers
cin >> val1 >> val2 >>val3;
cout <<val1<<" \n"<< val2<<"\n"<<
val3<<"\n";
cout << val3<<"\n"<< val2<<"\n"<<
  val1 << "\n";
return 0;
}
```

output (if input is 123)

1
2
3
3
2
1

```
C:\Users\PC\Documents\lab2_task3.exe
Please enter your 3 numbers:1 2 3
1
2
3
3
2
1

--------------------------------
Process exited after 6.351 seconds with return value 0
Press any key to continue . . .
```

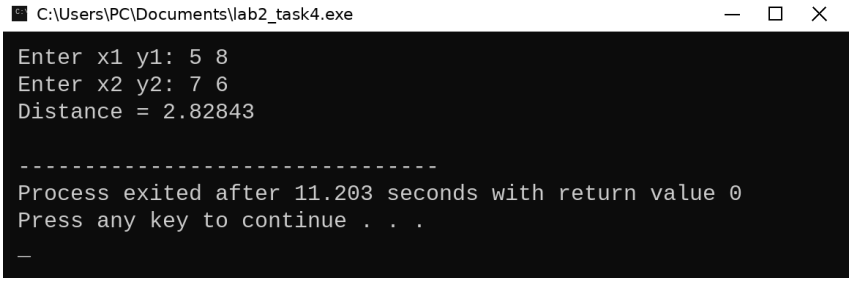
Task - 4

write c++ program to calculate distance b/w two points.

note x_1, y_1, x_2, y_3 are all double values

```
# include <iostream>
#include <<math
 using namespace std;
int main()
{
double x1, y1, x2, y2, d;
cout << "Enter x1 y1 : ";
 (in >> x1 >> y1;
cout << "Enter x2 y2 : ";
 (in >> x1 >> y1;
cout << "Enter x2 y2: ";
 (in >> x2 >> y2;
 d = Sqrt (Pow(x2 - x1, 2) + Pow(1/2 - y1, 2));
cout << " Distance = " << d << endl;
system ("pause");
 return 0;
}
```

Enter x_1 y_1 = 5 8
Enter x_2 y_2 = 7 6
Distance = 2.82843



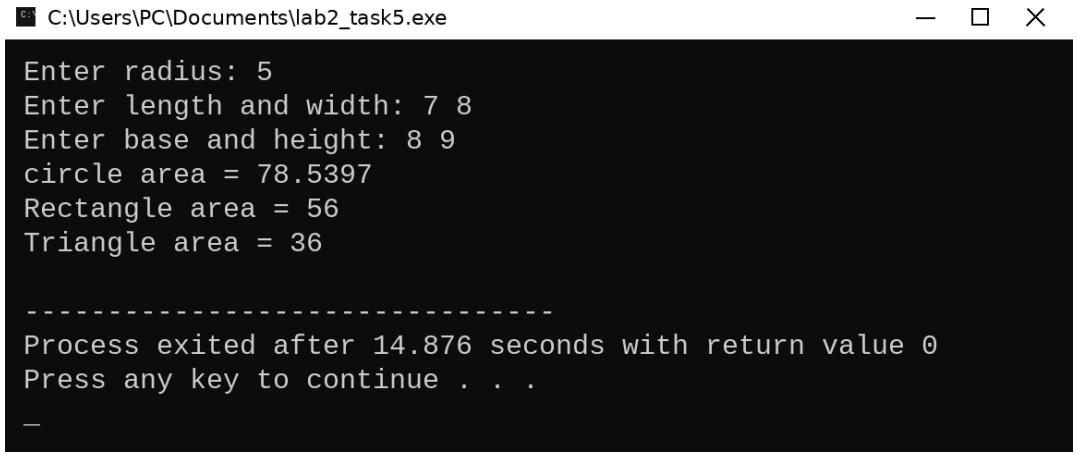
Task - 5

Find circle area, rectangle area
Triangle area

```
#include<iostream>
using namespace std;
int main()
{
cout double PI =3.14159;
double r, length, width, base, height
cout << "Enter radius: ";
cin >> r;
cout << "Enter length and width:";
cin >> length >> width;
cout << "Enter base and height:";
cin >> base >> height;
cout << "circle area:" << PI*r*r<<endl;
cout << "Rafagle area = " << length*width
<<endl;
```

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```
cout << "Triangle area = " << base
 * height * 0.5 << endl;
system ("Pause");
return 0;
}
Enter radius : 6
Enter length and width = 78
Enter base and height 89
circle area = 78.5397
Rectangle area = 42
Triangle = 32
```



Task 6

Total purchase. A customer in a store is purchasing five item's. price is given

⇒ price of item 1 = $ 12.95
price of item 2 = $ 24.95
Price of item 3 = $ 6.95
Price of item 4 = $ 14.95
Price of item 5 = $ 3.95

write a program that hold's that prices of the five item's in five variables. Display each item price the subtotal of the sale. The amount of sale, tax and total, Assume the sale's tax is 6%.

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```
#include <iostream>
using namespace std;
int main()
{
    double p1 = 12.95, p2 = 24.95, p3 = 6.95, p4 = 14.95
           p5 = 3.95;
    double Subtotal = p1+p2+p3+p4+p5;
    double tax = Subtotal * 0.06;
    double total = Subtotal + tax;
    cout << "item 1: $" << p1 << endl;
    cout << "item 2: $" << p2 << endl;
    cout << "item 3: $" << p3 << endl;
    cout << "item 4: $" << p4 << endl;
    cout << "item 5: $" << p5 << endl;
    cout << "Subtotal: $" << Subtotal << endl;
    cout << "Sales Tax (6%): $" << tax << endl;
    cout << "Total: $" << total << endl;
    System ("Pause");
    return 0;
}
```

item 1: $12.95
item 2: $ 24.95
item 3: $ 6.95
item 4: $ 14.95
item 5: $ 3.95
Subtotal: $ 63.75

Sales Tax(6%): 3.82$
Total: $ 67.57$

C:\Users\PC\Documents\lab2_task6.exe

```
item 1: $12.95
item 2: $24.95
item 3: $6.95
item 4: $14.95
item 5: $3.95
Subtotal: $63.75
Sales Tax(6%): $3.825
Total: $67.575

--------------------------------
Process exited after 0.04360 seconds with return value 0
Press any key to continue . . .
```

ay: Date:

Bonus task

Write a program that inputs a five-digit integer, Seprate the integer into individual digit and prints the digits space from each other by three space each.

```
#include <iostream>
using namespace std;
int main ()
{
  int n;
  cout << "Enter a 5-digit number:";
  cin >> n;
  cout <<n/10000 << "  "
      << (n/1000) %10 << "  "
      << (n/100) %10 << "  "
      << (n/10) %10 << "  "
      << n% << endl;
      System ("Pause");
      return 0;
    }
```

Enter a 5-digit number 24145

2 4 1 4 5

```
C:\Users\PC\Documents\lab2_bonus.exe
Enter a 5-digit number: 24145
2   4   1   4   5

--------------------------------
Process exited after 7.918 seconds with return value 0
Press any key to continue . . .
```

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Day: Date:

Home task 1

Swap

```
# include <iostream>
using namespace std;
int main ()
{
  int var1, var2, temp;
  cout << "Enter var1:";
  cin >> var1;
  cout << "Enter var2:";
  cin >> var2;
  temp = var1;
  var1 = var2;
  var2 =  temp;
  cout << "After Swap: var1: "<<var1<<";
          var2: "<< var2<< endl;
  System ("Pause");
  return 0;
```

output

```
Enter var1:10
Enter var2:20
After Swap: var1=20, var2: 10
```

```
C:\Users\PC\Documents\lab2_home1.exe
Enter var1: 10
Enter var2: 20
After Swap: var1: 20, var2: 10

--------------------------------
Process exited after 9.114 seconds with return value 0
Press any key to continue . . .
```

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Date:

Home task (2)

Swap two variable

```
#include <iostream>
using namespace std;
int main()
{
  char letter, upper;
  cout << "Enter an alphabet in Small caps
          (a-z):";
  cin >> letter;
  upper = letter - 32;
  cout << "upper case=" << upper << endl;
  System ("Pause");
  return 0;
}
```

```
Enter an alphabet in Small caps(a-z):
a b c d
upper case: A
```

